

Yeshwantrao Chavan College of engineering, Nagpur.

Data Analytics Lab 01

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Aim :	To study the basic Python tools used for data analytics and perform simple operations using NumPy and Pandas , such as creating arrays, creating Series and DataFrames, and viewing/sorting data.		
Implementation Steps :	Step 1: Import the required libraries Step 2: Create NumPy arrays Step 3: Display array properties Step 4: Create a Pandas Series Step 5: Create a Pandas DataFrame Step 6: Display and sort the data Step 7: Observe the output		
Program Code :	<pre># Lab 1: Introduction to Python Tools for Data Analytics # Step 1: Import libraries import numpy as np import pandas as pd import matplotlib.pyplot as plt print("----- NUMPY OPERATIONS -----") # Step 2: Create a 1D NumPy array data1 = [6, 7.5, 8, 0, 1] arr1 = np.array(data1) print("1D Array:") print(arr1) # Step 3: Create a 2D NumPy array data2 = [[1, 2, 3, 4], [5, 6, 7, 8]] arr2 = np.array(data2) print("\n2D Array:") print(arr2) # Step 4: Array properties print("\nNumber of dimensions:", arr2.ndim) print("Shape of array:", arr2.shape) print("Size of array:", arr2.size) print("Data type of elements:", arr2.dtype) print("\n----- PANDAS OPERATIONS -----") # Step 5: Create a Pandas Series s = pd.Series([1, 3, 5, np.nan, 6, 8])</pre>		

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print("\nPandas Series:")
print(s)

# Step 6: Create date index
dates = pd.date_range("20130101", periods=6)
print("\nDate Range:")
print(dates)

# Step 7: Create DataFrame using random numbers
df = pd.DataFrame(np.random.randn(6, 4), index=dates, columns=list("ABCD"))
print("\nDataFrame with random values:")
print(df)

# Step 8: Create DataFrame from dictionary
data = {
    "Name": ["Amit", "Neha", "Ravi"],
    "Age": [20, 21, 19],
    "City": ["Nagpur", "Pune", "Mumbai"]
}
df2 = pd.DataFrame(data)
print("\nDataFrame from dictionary:")
print(df2)

# Step 9: Display first 5 rows
print("\nFirst 5 rows of df:")
print(df.head())

# Step 10: Display last 3 rows
print("\nLast 3 rows of df:")
print(df.tail(3))

# Step 11: Display data types
print("\nData types of df:")
print(df.dtypes)

# Step 12: Sort values by column B
print("\nSorted DataFrame by column B:")
print(df.sort_values(by="B"))
```

Output :

	PassengerId	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	zero.4	...	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	Embarked	zero.17	zero.18	Survived
0	1	22.0	7.2500	0	1	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1	2	38.0	71.2833	1	1	0	0	0	0	0	...	0	0	0	1	0	0	0.0	0	0	1
2	3	26.0	7.9250	1	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	1
3	4	35.0	53.1000	1	1	0	0	0	0	0	...	0	0	0	1	0	0	2.0	0	0	1
4	5	35.0	8.0500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1304	1305	28.0	8.0500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1305	1306	39.0	108.9000	1	0	0	0	0	0	0	...	0	0	0	1	0	0	0.0	0	0	0
1306	1307	38.5	7.2500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1307	1308	28.0	8.0500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1308	1309	28.0	22.3583	0	1	0	0	0	0	0	...	0	0	0	3	0	0	0.0	0	0	0

300 rows x 22 columns

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	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	zero.4	...	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	Embarked	zero.17	zero.18	Survived
0	1	22.0	7.2500	0	1	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1	2	38.0	71.2833	1	1	0	0	0	0	0	...	0	0	0	1	0	0	0.0	0	0	1
2	3	26.0	7.9250	1	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	1
3	4	35.0	53.1000	1	1	0	0	0	0	0	...	0	0	0	1	0	0	2.0	0	0	1
4	5	35.0	8.0500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1304	1305	28.0	8.0500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1305	1306	39.0	108.9000	1	0	0	0	0	0	0	...	0	0	0	1	0	0	0.0	0	0	0
1306	1307	38.5	7.2500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1307	1308	28.0	8.0500	0	0	0	0	0	0	0	...	0	0	0	3	0	0	2.0	0	0	0
1308	1309	28.0	22.3583	0	1	0	0	0	0	0	...	0	0	0	3	0	0	0.0	0	0	0

Result :	Thus, the basic Python tools for data analytics were studied successfully. NumPy arrays and Pandas Series/DataFrames were created, and basic operations such as viewing data, checking dimensions, and sorting values were performed successfully.
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Marks :		Name of Lecturer : Dr. S.A. Dhondse
		Signature :